ASTANA IT UNIVERSITY **HIGHER EDUCATION PEDAGOGY** • MASTER'S DEGREE PROGRAM • DEPARTMENT OF GENERAL EDUCATION DISCIPLINES

FINAL CRITICAL ANALYSIS POSTER PRESENTATION • GROUP SSE-2601

THE ICEBERG OF LEARNING

What Lies Beneath a High Grade?

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PRIMARY RESEARCH QUESTION

"Do high grades necessarily reflect actual understanding and the ability to apply knowledge?"

Focus: Evaluating the divergence between surface test metrics and multidimensional cognitive mastery.

SECTION 1 • INTRODUCTION & PEDAGOGICAL CONTEXT

PEDAGOGICAL SIGNIFICANCE

INTRODUCTORY LEARNING HOOK (HYPOTHETICAL SCENARIO)

"You score 95/100. But can you still explain what you learned next week?"

Hypothetical learning scenario reflecting common educational experience: high exam marks often correlate with cramming rather than durable mastery.

Grades remain central to higher education governance, serving as credentials for progression, honors, and employment. However, high examination scores often reflect temporary memorization strategies rather than durable conceptual mastery.

SSE-2601 RESEARCH THESIS:

"Grades are useful indicators of academic performance, but they cannot independently demonstrate the full depth of student learning."

JOHN BIGGS (1996)

Constructive Alignment

Intended Learning Outcomes (ILOs), teaching-learning activities (TLAs), and assessment tasks must form an integrated, coherent system.

CONSTRUCTIVE ALIGNMENT ARCHITECTURE — TRIPARTITE LINKAGE

INTENDED OUTCOMES	TEACHING ACTIVITIES	ASSESSMENT METHODS
High-order mastery	Active problem-solving	Authentic application

CRITICAL PEDAGOGICAL QUESTION:

"Are we assessing what we actually intended to teach?"

PAULO FREIRE (2000)

Banking vs. Problem-Posing

Critiques the 'banking model' where students act as passive containers of deposited facts. Champions problem-posing dialogue and active co-construction.

BANKING MODEL	PROBLEM-POSING
Passive fact deposition & recall	Critical dialogue & co-creation

CRITICAL PEDAGOGICAL QUESTION:

"Are students passive recipients or active co-creators?"

FORMATIVE FEEDBACK PRINCIPLE

ASSESSMENT AS LEARNING

Assessment must serve as a continuous learning instrument rather than a retroactive post-mortem grade.

ASSESS	FEEDBACK	REFLECT	IMPROVE
Diagnostic drafts, peer reviews, low-stakes tests	Timely, specific, criteria-aligned guidance	Student self-interrogation & gap identification	Concrete revision before summative judgment

"Feedback is only valuable when students are afforded the structured opportunity to interpret and apply it."

SURFACE CRITERIA • WHAT WE CAN SEE

LEVEL 01 • VISIBLE SURFACE • 0 m

ACADEMIC PERFORMANCE

- Examination scores & test points
- Correct formulaic answers
- Reproduced textbook definitions
- Completed assessment requirements

The visible grade provides quantifiable evidence of performance in a specific, timed assessment task. However, it does not automatically reveal how deeply or flexibly the student understands underlying principles.

💡 Essential for baseline benchmarking, but only represents the tip of cognitive capability.

LAYER 01 • MEANING-MAKING

LEVEL 02 • SUBMERGED DEPTH • -150 m

CONCEPTUAL UNDERSTANDING

- Explain ideas in your own words
- Connect related & disparate concepts
- Understand why mechanisms function
- Recognize principles across domains

Meaningful learning transcends rote recitation. Students grasp structural relationships and can explain the foundational rationale without relying on memorized scripts.

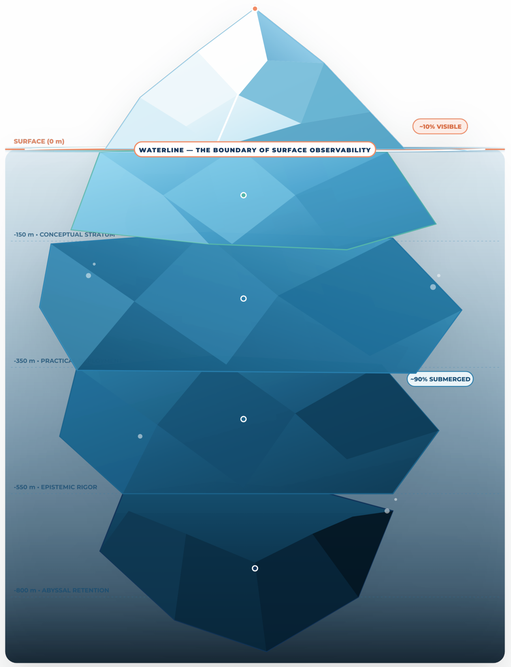
LAYER 03 • EPISTEMIC RIGOR

LEVEL 04 • SUBMERGED DEPTH • -550 m

CRITICAL THINKING

- Question foundational assumptions
- Critically evaluate alternatives
- Defend methodological decisions
- Synthesize independent viewpoints

Students actively interrogate ideas, critique methodologies, and adapt to ambiguity instead of treating knowledge as a static repository of unquestioned truths.



LAYER 02 • ACTIVE DEPLOYMENT

LEVEL 03 • SUBMERGED DEPTH • -350 m

PRACTICAL APPLICATION

- Apply theory to unfamiliar problems
- Solve realistic engineering tasks
- Justify chosen design trade-offs
- Diagnose system errors & anomalies

Mastery becomes visible when students deploy theoretical knowledge to build, analyse, and troubleshoot real systems. Illustrated in Computer Architecture: understanding CPU ↔ RAM bus interactions rather than just defining registers.

💡 Contextual problem-solving validates genuine competence.

LAYER 04 • DURABLE RETENTION

LEVEL 05 • SUBMERGED DEPTH • -800 m

LONG-TERM RETENTION

- Durable recall long after exam week
- Explain concepts spontaneously without notes
- Anchor new knowledge to existing schema
- Transfer insights into professional practice

The true measure of learning is durability. Meaningful knowledge integrates into the student's cognitive framework, remaining accessible and actionable across semesters and careers.

Conceptual Principle:
These five dimensions represent complementary aspects of cognitive depth, not a rigid sequential hierarchy. Grades reveal localized performance; authentic learning requires conceptual, practical, and epistemological roots.

CRITICAL ANALYSIS: EXAMINATIONS VS. PROJECTS

SHOWING BOTH SIDES

TRADITIONAL EXAMINATIONS

STRENGTHS:
Efficient verification of foundational terminology & core facts

LIMITATIONS:
Prone to temporary cram-and-forget memorization strategies

PRACTICAL & AUTHENTIC PROJECTS

STRENGTHS:
Direct demonstration of theoretical knowledge applied to real tools

LIMITATIONS:
Project completion does not automatically prove individual mastery

Balanced Consensus: *"Neither examination scores nor completed projects provide a complete picture of learning when considered in isolation. True rigor emerges from their deliberate combination."*

GRABINGER & DUNLAP (1995)

Rich Environments for Active Learning (REAL)

Promotes learning through authentic contexts, collaborative responsibility, dynamic problem-solving, and reflective inquiry.

CRITICAL PEDAGOGICAL QUESTION:

"Can students utilize knowledge outside the textbook sandbox?"

SECTION 3 • HE APPLICATION

BALANCED ASSESSMENT ECOSYSTEM

5 Complementary Modalities

Examinations	Practical Projects	Case Studies	Presentations & Q&A	Formative Feedback
Verify core theoretical knowledge, individual mastery, and fluency in basic terminology under controlled conditions.	Deploy principles in realistic environments, evaluate system trade-offs, and produce authentic artifacts.	Analyze complex, ill-structured dilemmas, balance competing engineering priorities, and justify solutions.	Articulate concepts verbally, respond to critical scrutiny, and defend technical decisions in real time.	Provide timely diagnostic feed-forward before high-stakes milestones, turning assessment into a learning tool.

COMPUTING SYSTEMS ARCHITECTURE APPLICATION — SSE-2601 Case

CPU (PROCESSING)	Bus Interaction	RAM (MEMORY)
Registers & Control ALU		Address Bus & Latency

"Students demonstrate true understanding when they can explain system behaviour and justify design decisions, not just recite hardware specs."

EVIDENCE-INFORMED HE RECOMMENDATIONS:

01. **Constructive Alignment:** Align examination prompts and practical rubrics directly with intended higher-order learning outcomes.
02. **Dual Assessment Modality:** Combine timed individual verification with authentic practical tasks and oral defenses.
03. **Active Dialogue & Defense:** Integrate peer feedback, classroom debate, and verbal justification into technical coursework.
04. **Actionable Formative Loops:** Ensure draft milestones receive timely, actionable feedback before final summative grading.

✓ **Measure not only what students can reproduce, but also what they can explain, apply, and retain.**

SECTION 4 • FINAL CONCLUSION

SSE-2601 Consensus

A high grade may indicate successful preparation, but it is not sufficient proof of deep understanding. Meaningful learning becomes visible when students can explain ideas, apply knowledge in unfamiliar situations, evaluate their solutions, and retain what they have learned across time.

"GRADES SHOULD MEASURE LEARNING — NOT DEFINE ITS LIMITS."

SECTION 5 • REFERENCES (APA 7TH EDITION) & RESEARCH CONTRIBUTIONS

Verified Scholarly Literature

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INDIVIDUAL RESEARCH CONTRIBUTIONS (SSE-2601)

Nikita Shulmin
Academic performance, memorisation-based assessment, practical assessment, and feedback.

Dana Kenzhebayeva
Constructive alignment, practical judgment, educational philosophy, and the balance between structure and flexibility.

Islam Akhmetov
Practical learning, formative feedback, and theoretical knowledge applied to Computer Organisation and Architecture.

Abzhappar Abylaikhan
Interactive teaching, student participation, classroom discussions, and group-based mini-lessons.